Experiment 3

Design of Test Cases and a Maintenance Plan for a Sample Application

Sample Application: Hospital Management System

Aim

The aim of this experiment is to design test cases and prepare a maintenance plan for a Hospital Management System. Test cases are used to verify whether each module of the system functions correctly according to the specified requirements. The maintenance plan ensures that the software continues to perform efficiently after deployment by fixing bugs, improving security, and updating features. This experiment helps understand the importance of software testing and maintenance in delivering reliable healthcare software.

Objectives

- The objectives of this experiment are:
- To understand the importance of software testing.
- To design effective test cases for different modules.
- To verify that the application meets user requirements.
- To identify defects before software deployment.
- To prepare a maintenance plan for long-term software support.
- To improve software quality, reliability, and performance.

Introduction

A Hospital Management System (HMS) is a software application used to manage the daily operations of a hospital. It stores patient information, schedules appointments, manages doctors, processes billing, and maintains medical records. Since hospitals depend on accurate and secure information, the software must be thoroughly tested

before deployment. Regular maintenance is also essential to ensure continuous availability, improve performance, and protect sensitive patient data.

About the Hospital Management System

The Hospital Management System integrates various hospital activities into a single platform. Patients can register, book appointments, and access their medical records. Doctors can manage patient information, prescribe treatments, and update reports. Administrators can monitor hospital operations, billing, staff records, and inventory. The system improves efficiency, reduces paperwork, and ensures quick access to healthcare information.

Testing Objectives

The testing process aims to ensure that all modules of the Hospital Management System function correctly.

- ❖ Verify patient registration and login.
- ❖ Test appointment booking and cancellation.
- ❖ Check doctor and staff management features.
- ❖ Validate billing and payment calculations.
- ❖ Ensure medical records are stored securely.
- ❖ Confirm report generation and search functions work properly.
- ❖ Verify system performance under multiple users.
- ❖ Ensure data security and privacy.

Test Cases

Test Case ID	Module	Test Scenario	Expected Result	Status
TC01	Patient Registration	Enter valid patient details	Registration successful	Pass
TC02	Login	Enter valid username and	Login successful	Pass

		password		
TC03	Appointment	Book appointment with doctor	Appointment confirmed	Pass
TC04	Billing	Generate patient bill	Correct bill generated	Pass
TC05	Medical Records	Search patient record	Patient record displayed	Pass
TC06	Logout	Click logout button	User logged out successfully	Pass

Types of Testing

The Hospital Management System undergoes different types of testing to ensure software quality.

- **Functional Testing** – Verifies all hospital modules work correctly.
- **Integration Testing** – Ensures communication between modules.
- **System Testing** – Tests the complete hospital application.
- **Performance Testing** – Checks response time during heavy usage.
- **Security Testing** – Protects confidential patient information.
- **User Acceptance Testing** – Confirms the software meets hospital requirements.

Defect Reporting

- Whenever a defect is identified during testing, it is documented for correction.
- A defect report includes:
- Defect ID
- Module Name
- Description of the Issue
- Steps to Reproduce
- Expected Result
- Actual Result
- Severity (Low/Medium/High)
- Current Status

- ❖ Proper defect reporting helps developers resolve issues efficiently before software deployment.

Maintenance Plan

- ❖ After deployment, the Hospital Management System requires continuous maintenance to ensure reliable operation.
- ❖ The maintenance activities include:
- ❖ Fix software bugs reported by users.
- ❖ Update patient and hospital management features.
- ❖ Improve system security with regular patches.
- ❖ Monitor server and database performance.
- ❖ Backup hospital records regularly.
- ❖ Improve software performance based on user feedback.
- ❖ Ensure compatibility with new operating systems.
- ❖ Provide technical support to hospital staff.

Types of Software Maintenance

Software maintenance is classified into four major categories.

Corrective Maintenance

- ❖ Fixes software bugs and system errors.
- ❖ Resolves issues reported by users.
- ❖ Improves system stability.

Adaptive Maintenance

- ❖ Updates the software to support new technologies.
- ❖ Ensures compatibility with updated hardware and operating systems.
- ❖ Modifies the application based on changing hospital requirements.

Perfective Maintenance

- ❖ Adds new features requested by hospitals.
- ❖ Improves user interface and overall performance.

- ❖ Enhances reporting and data management.

Preventive Maintenance

- ❖ Identifies possible future issues.
- ❖ Optimizes system performance.
- ❖ Updates documentation and improves code quality.

Results

The designed test cases successfully verify the major functions of the Hospital Management System, including patient registration, appointment booking, billing, and medical record management. The maintenance plan provides a structured approach for fixing defects, improving performance, and updating the application. Proper testing and maintenance help ensure reliable, secure, and efficient hospital operations.

Conclusion

This experiment demonstrates the importance of designing effective test cases and implementing a maintenance plan for the Hospital Management System. Testing ensures that all modules function correctly before deployment, while maintenance keeps the software secure, updated, and efficient after deployment. Together, they improve software quality, increase user satisfaction, and support the smooth operation of hospital services.